TM354 Block 2
From analysis to design

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Plan

- Review activity diagrams from ShareSpace
- Review unit 5
- Get an overview of the topics covered in units 6-7

Make sure you get the primers from the tutor group forum, and annotate them as we go along!

Block 1

BLOCK 1 REVIEW

Block 1: Your activity diagrams

- We will
  - Discuss how the diagrams evolved
  - Identify any key errors remaining
  - Discuss how different approaches help understand the problem

Block 2

UNIT 5 REVIEW

Question 1

Candidate concepts can be found by searching for nouns (p8), work through the Walton Hire statement and identify the candidate concepts (p198).

Walton hire

Walton Hire operates two types of membership scheme: standard membership and special membership. For both types of membership, a person registers with the shop by filling in a form and providing reliable proof of identity and gets a membership card with a unique member code. All members may enter into loan agreements for up to three items for a 24-hour period. For an extra fee, all members may borrow up to three items from any time after 6 pm on Friday to before 6 pm the following Monday. Standard members are not allowed to extend the loan period…

Question 2

Discuss how we could reduce this list to a set of key concepts. (p9/p198)

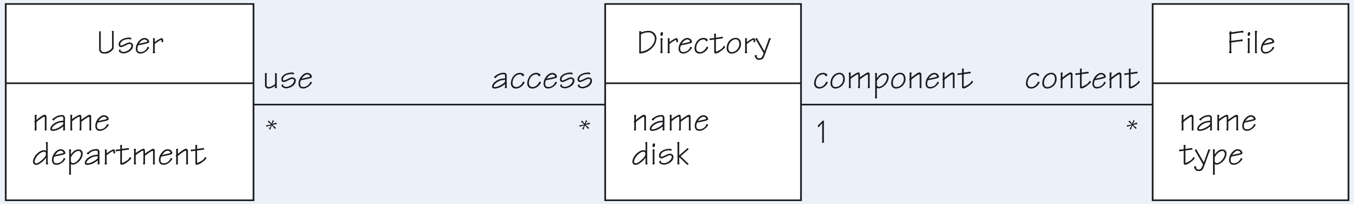
Question 3

Consider the classes to represent the concepts you have proposed and describe these classes and their relationships change over time. Identify the kinds of things we need to record. (see p11)

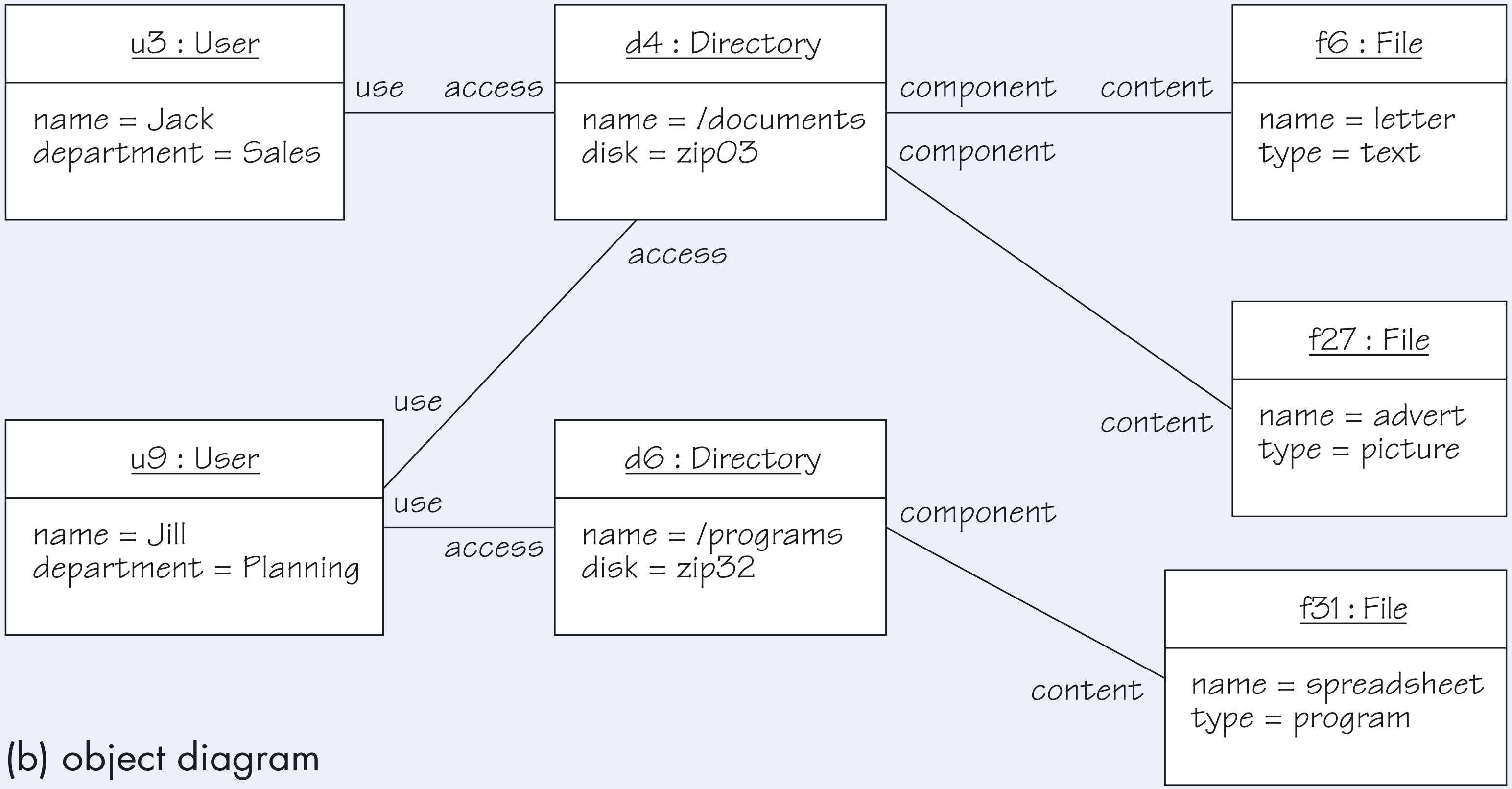
Question 4

What should we model? Does our approach (Plan driven/ Agile) change this? (p7)

- Note figure numbers on the following slides may not match the figure numbers in the units in all cases



(a) class diagram



(b) object diagram

Multiple associations

Time as a factor

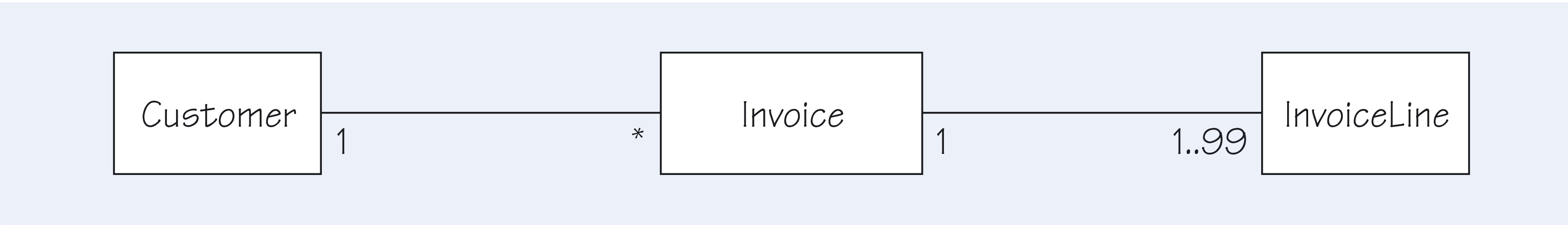


Figure 18 Invoices

This might look sensible, as a company would not normally send out an invoice that had no lines. However, if you consider the entire life history of an invoice, you may discover that there are circumstances in which it contains no invoice lines. An invoice may be created when an order is placed, but the invoice lines are only populated as the items are picked or manufactured. It could be that a zero-value invoice is created when an account is closed. Similarly, during editing, all lines might be deleted before new lines are added, and the invoice may remain in this state for some days. It seems that the model is probably wrong.

Composition and Aggregation

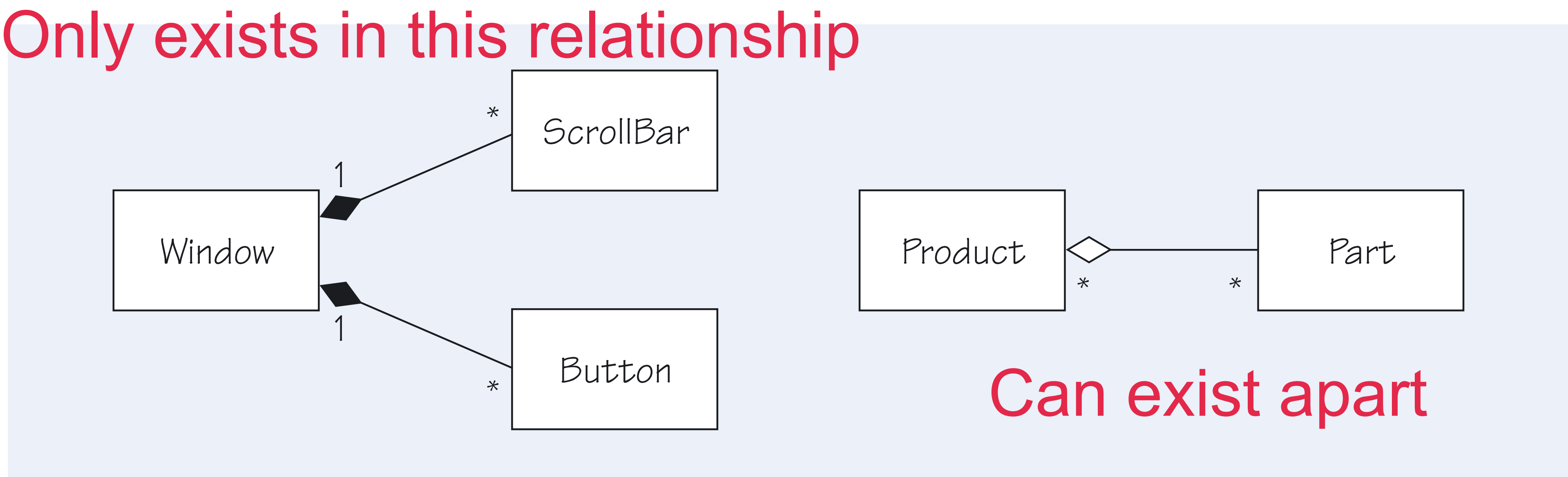


Figure 21 Composition and aggregation notation

Qualified associations

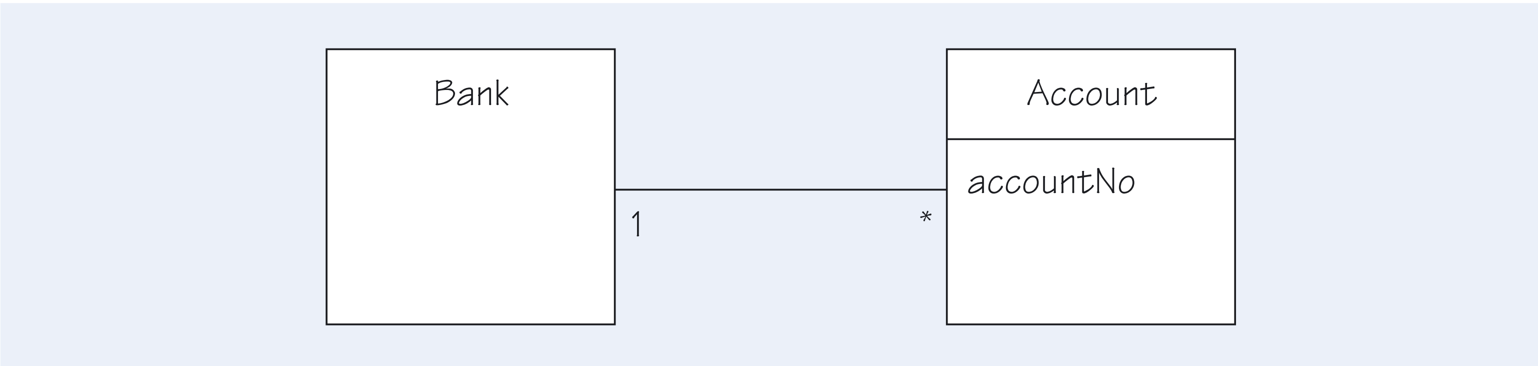


Figure 24 Account with account number

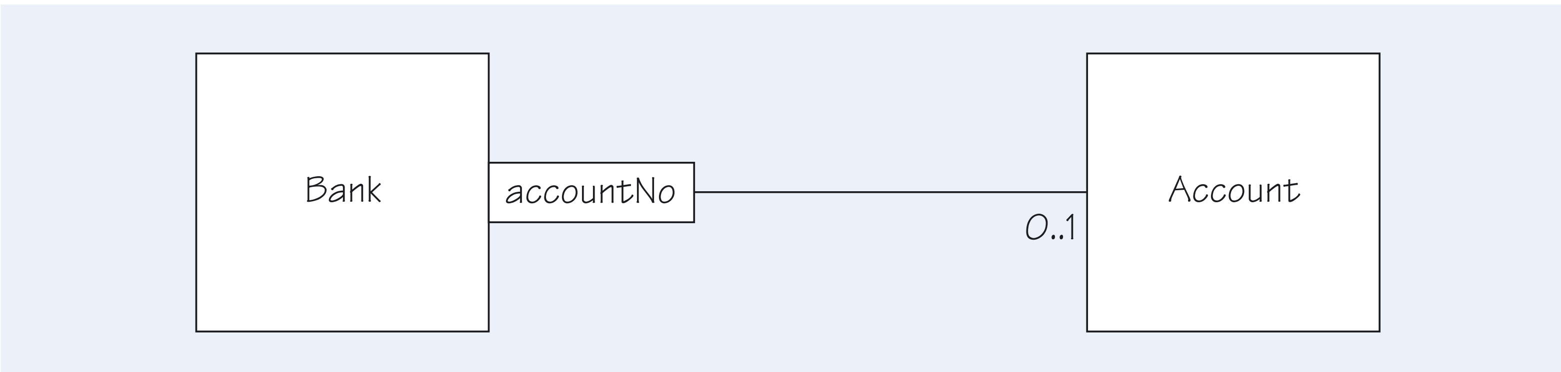


Figure 25 Using a qualified association

Specialization

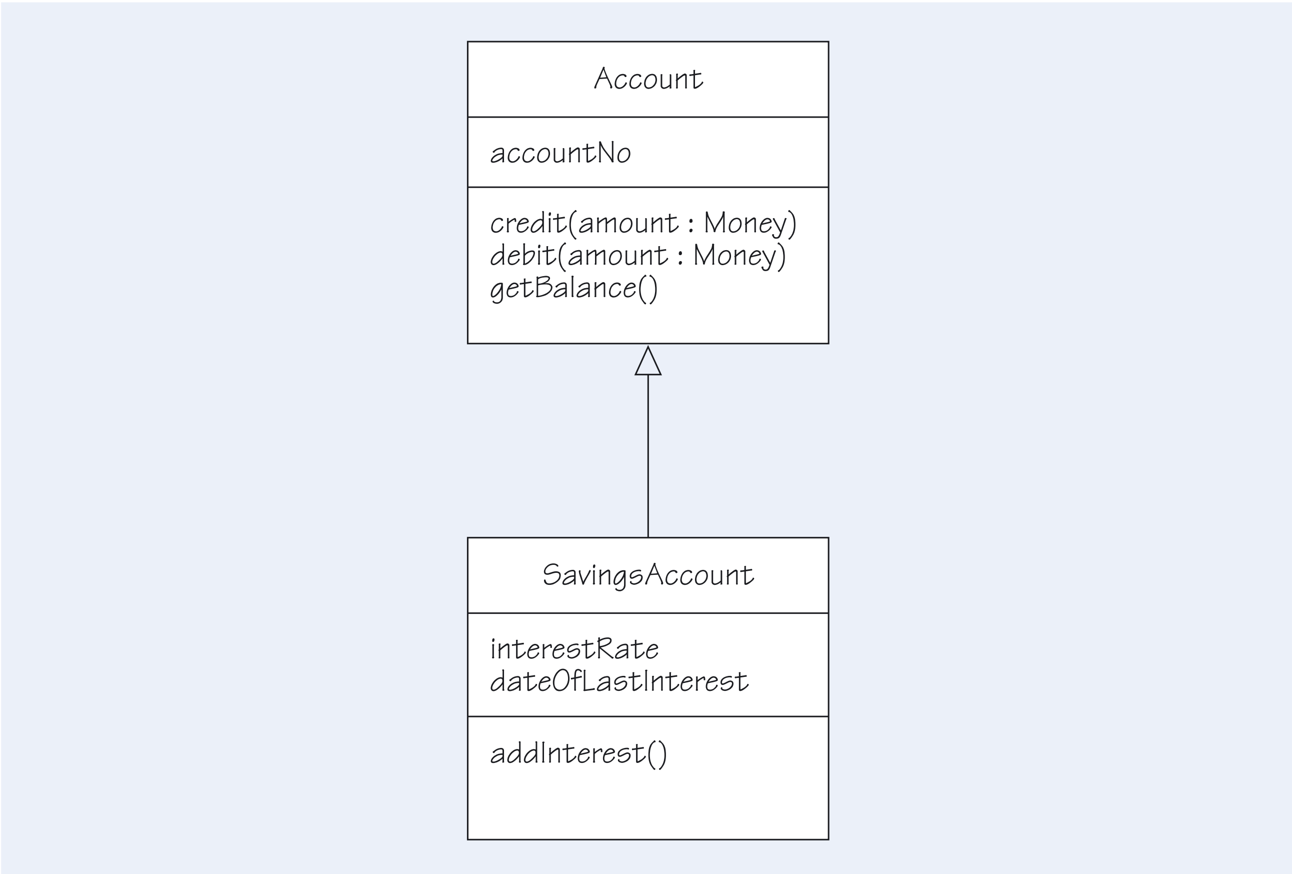


Figure 30 Specialisation of *Account*

Constraints

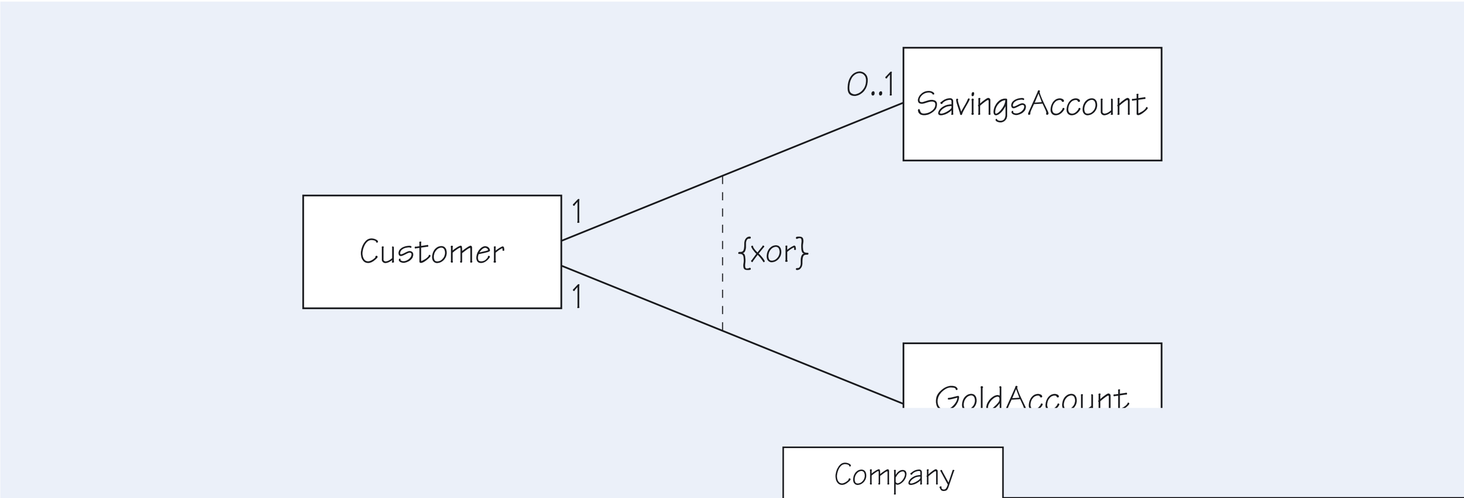


Figure 45 Controlling the use of d

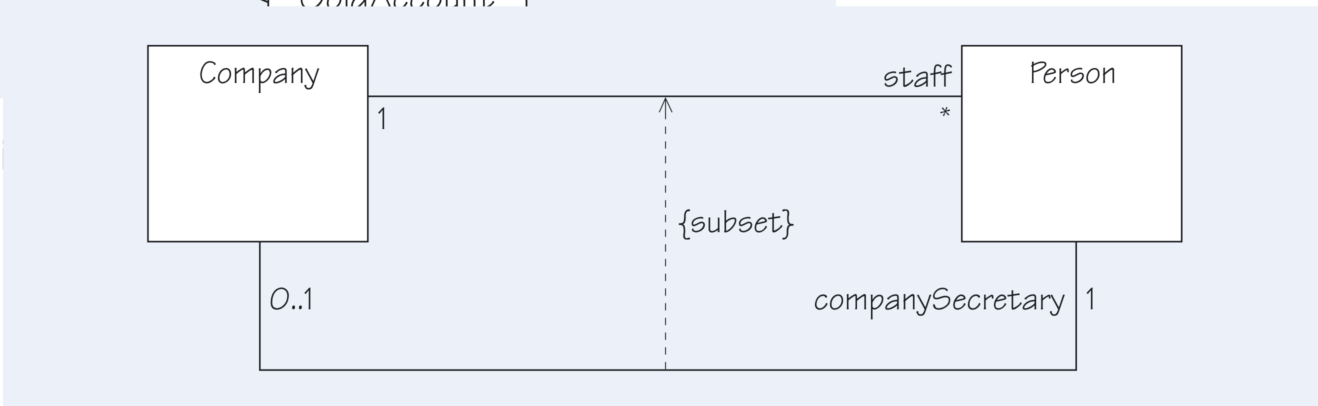


Figure 43 Formal constraints

Object Constraint Language

context Guest **inv**:

self.currentRoom.notEmpty() **implies** self.currentRoom.hotel = self.hotelReservation

context Committee **inv**:

self.chair <> self.secretary

Block 2

LOOKING AHEAD, UNITS 6-7

Unit 6

Design by Contract

- Preconditions
- Post conditions
- Behavior

Sequence diagrams

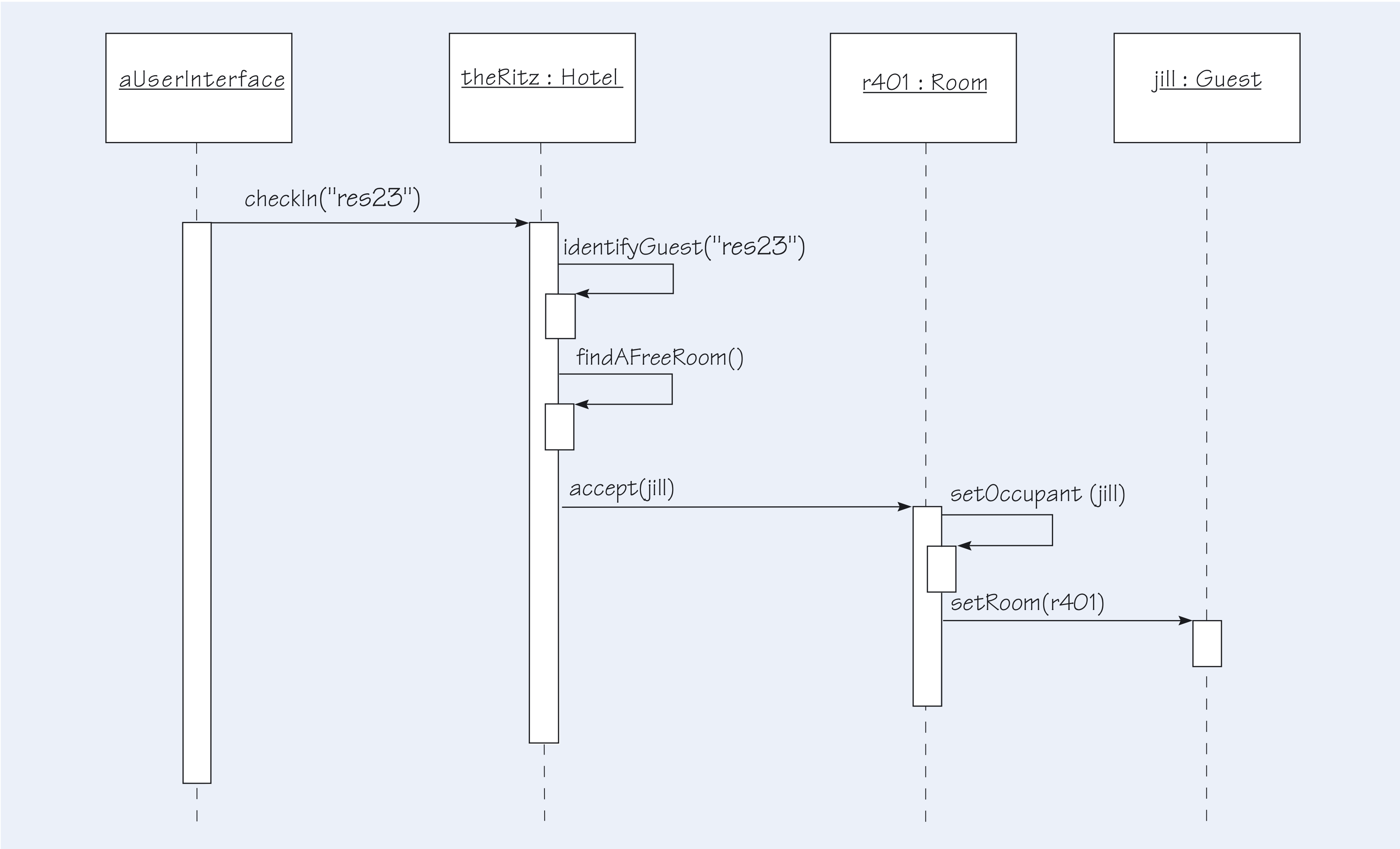


Figure 6 One way to check a guest in to a room

Communication diagram

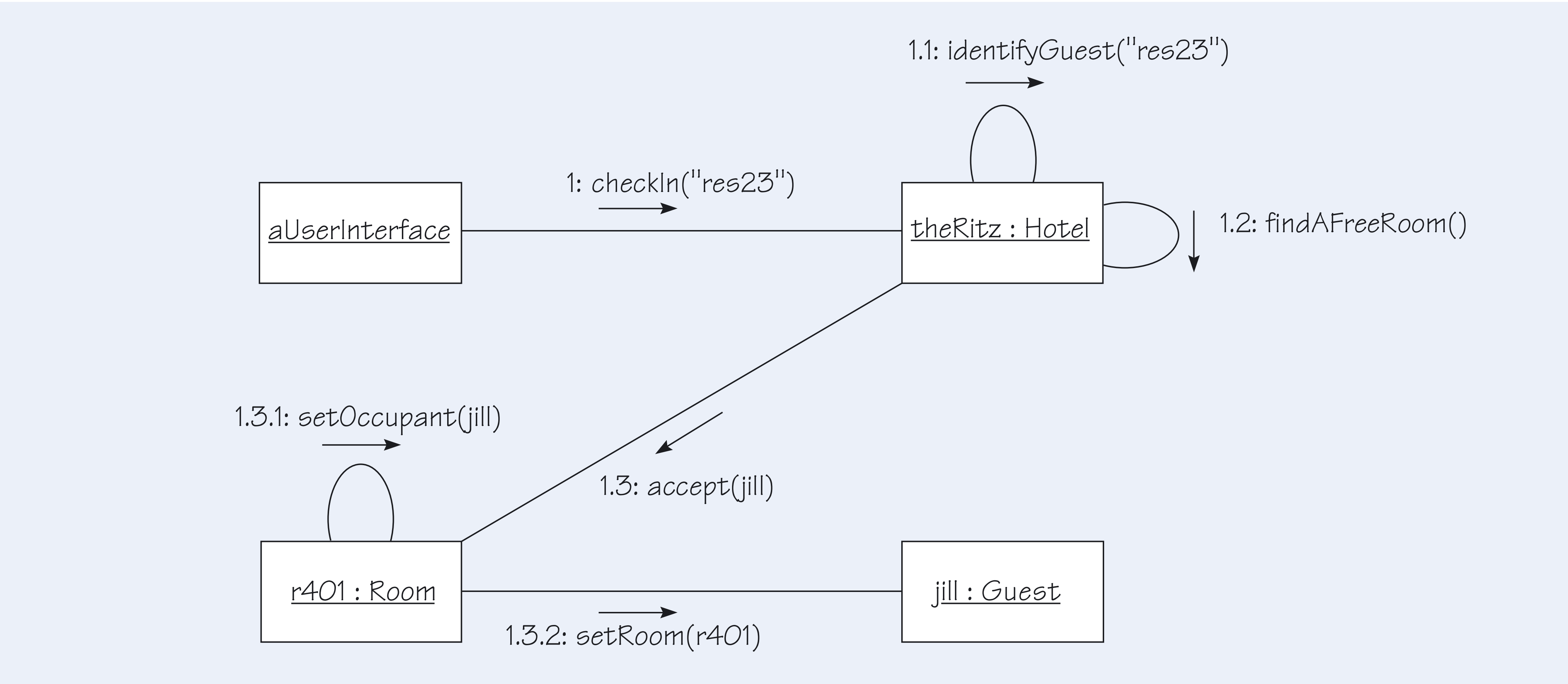


Figure 9 An equivalent communication diagram to Figure 6

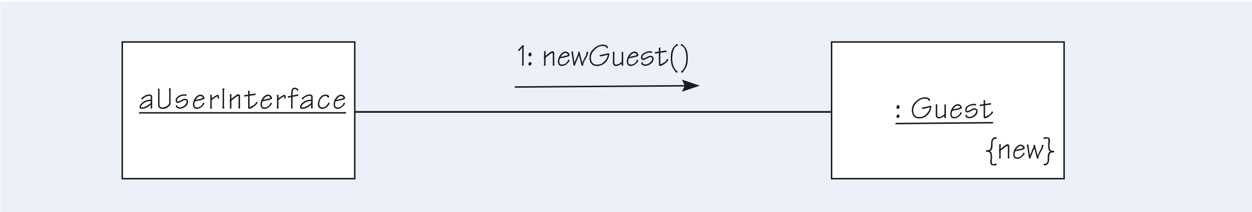


Figure 14 Object creation in a communication diagram

Creation and deletion

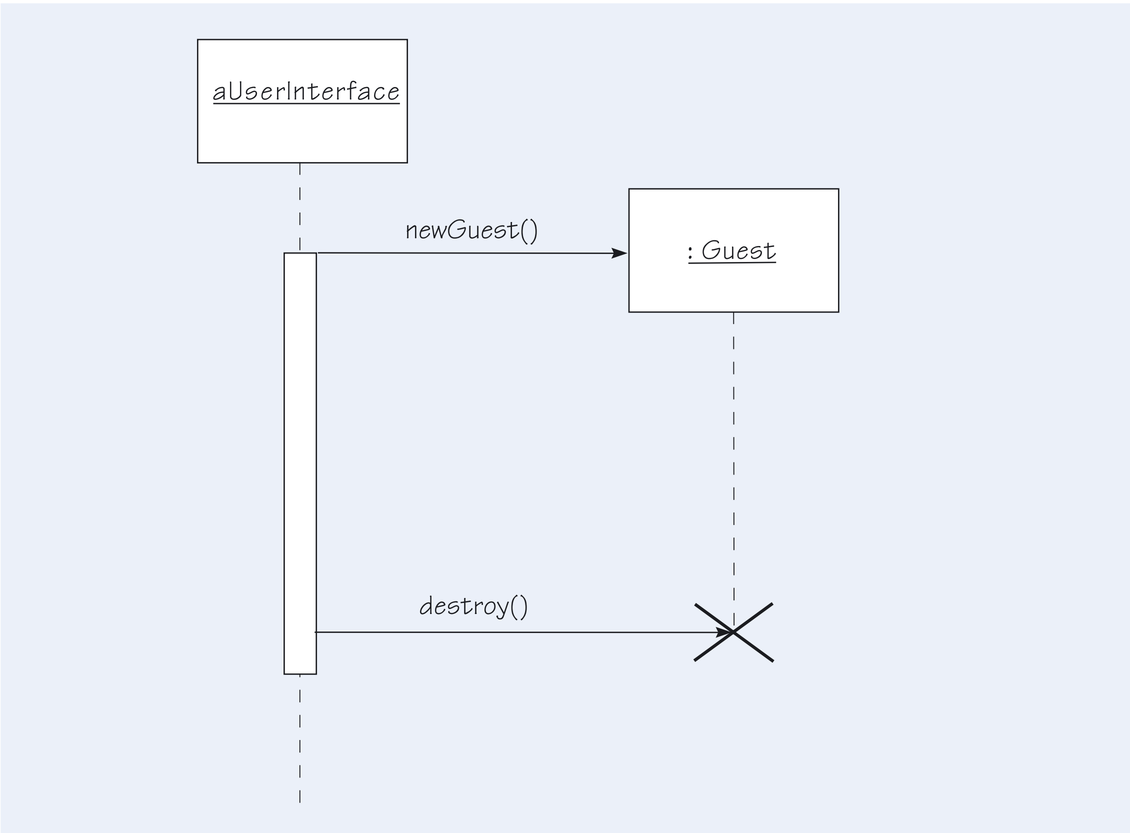


Figure 13 Creating and destroying an object in a sequence diagram

Alternative paths (unit 7)

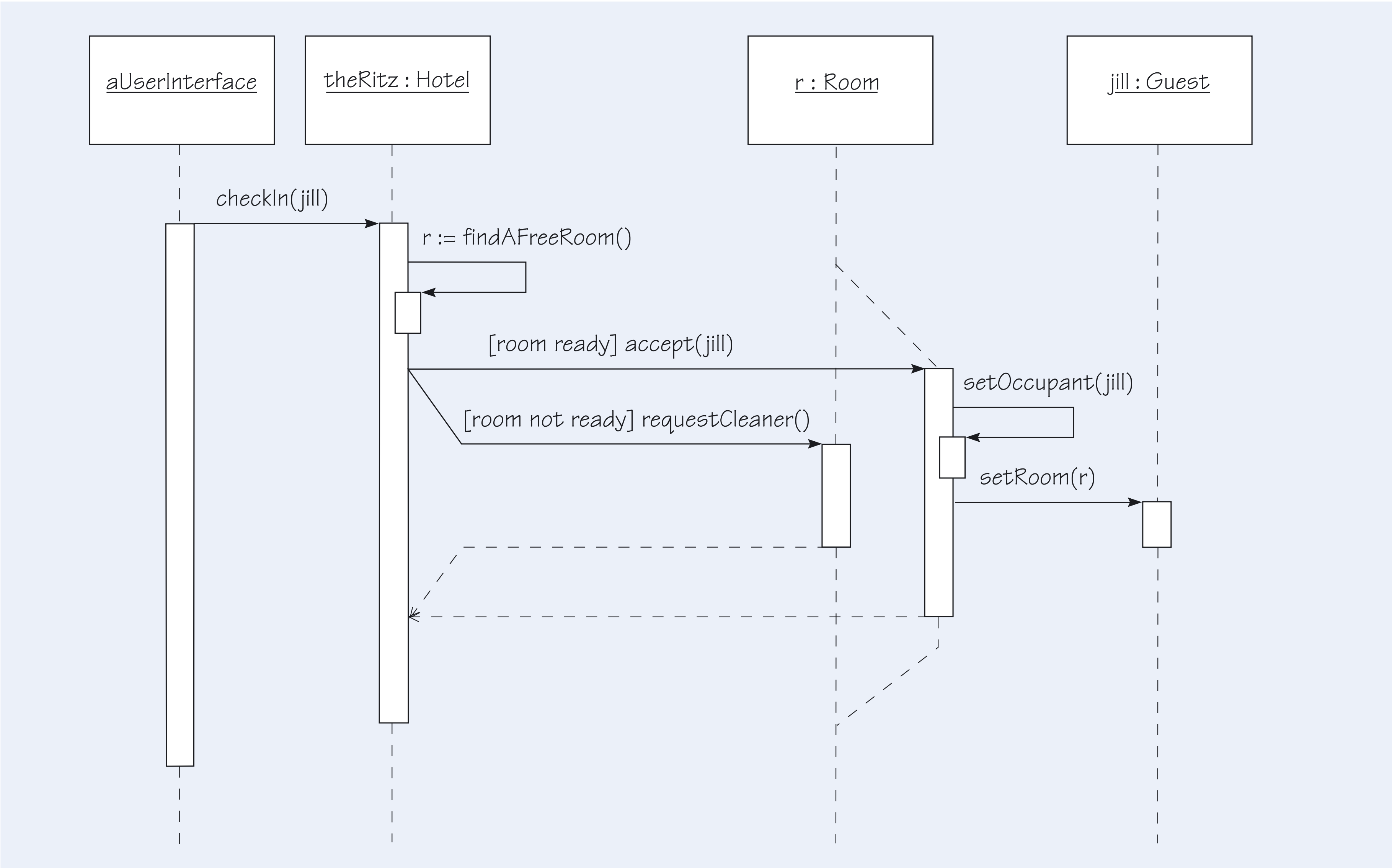


Figure 2 Alternative course of action in a sequence diagram

Iteration (unit 7)

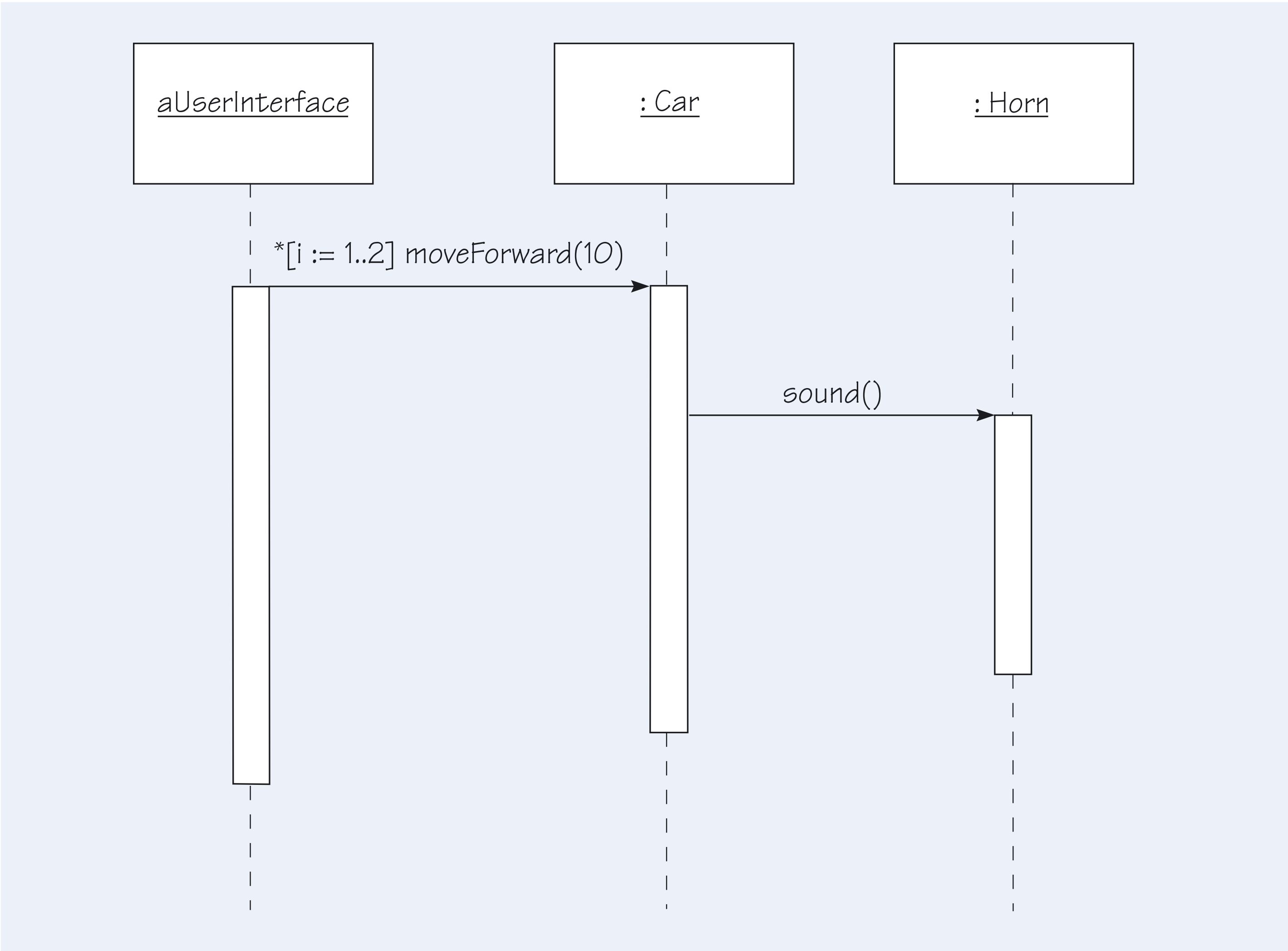


Figure 3 A sequence diagram showing iteration

Unit 7

State Machines

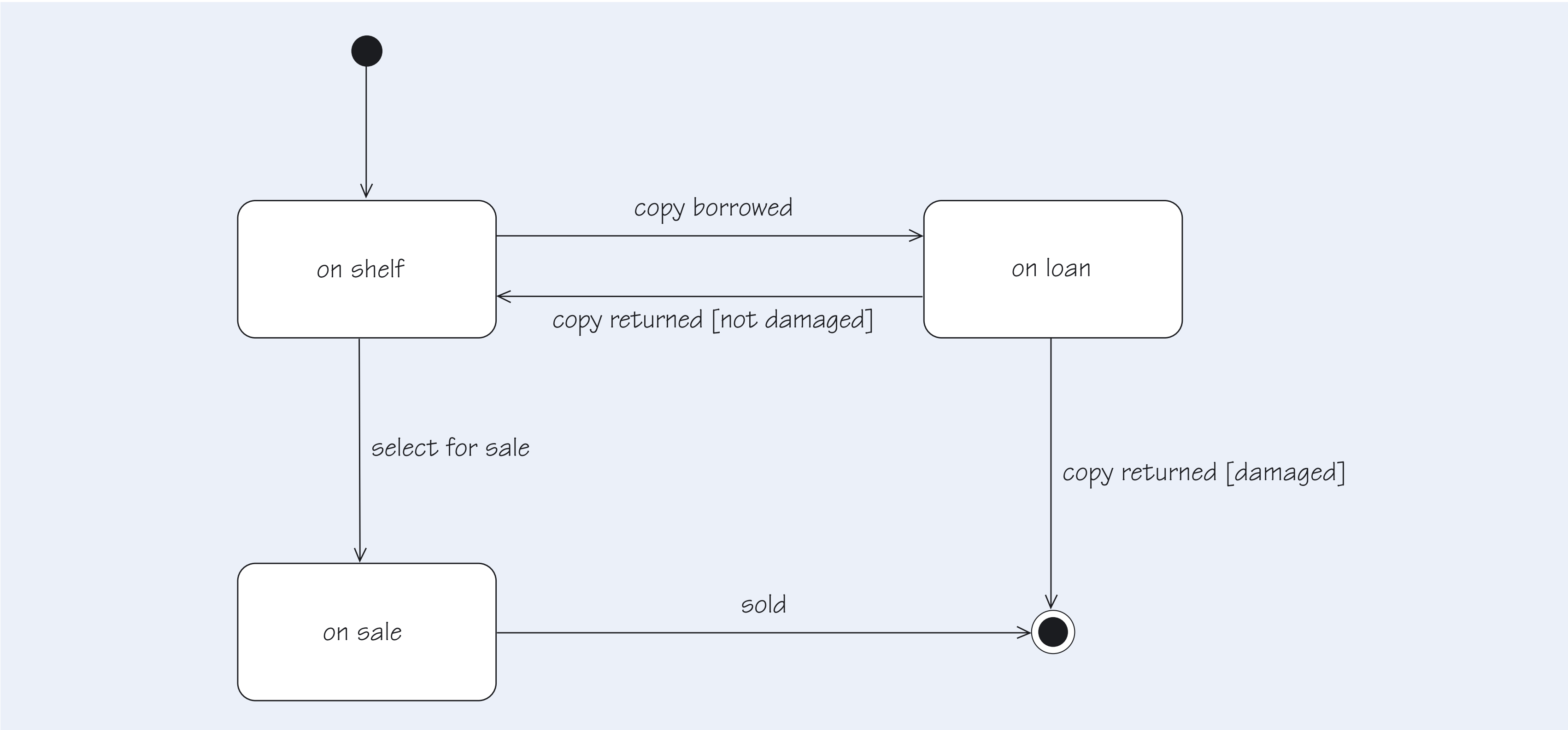


Figure 22 The life history of a *Copy* object

Packages

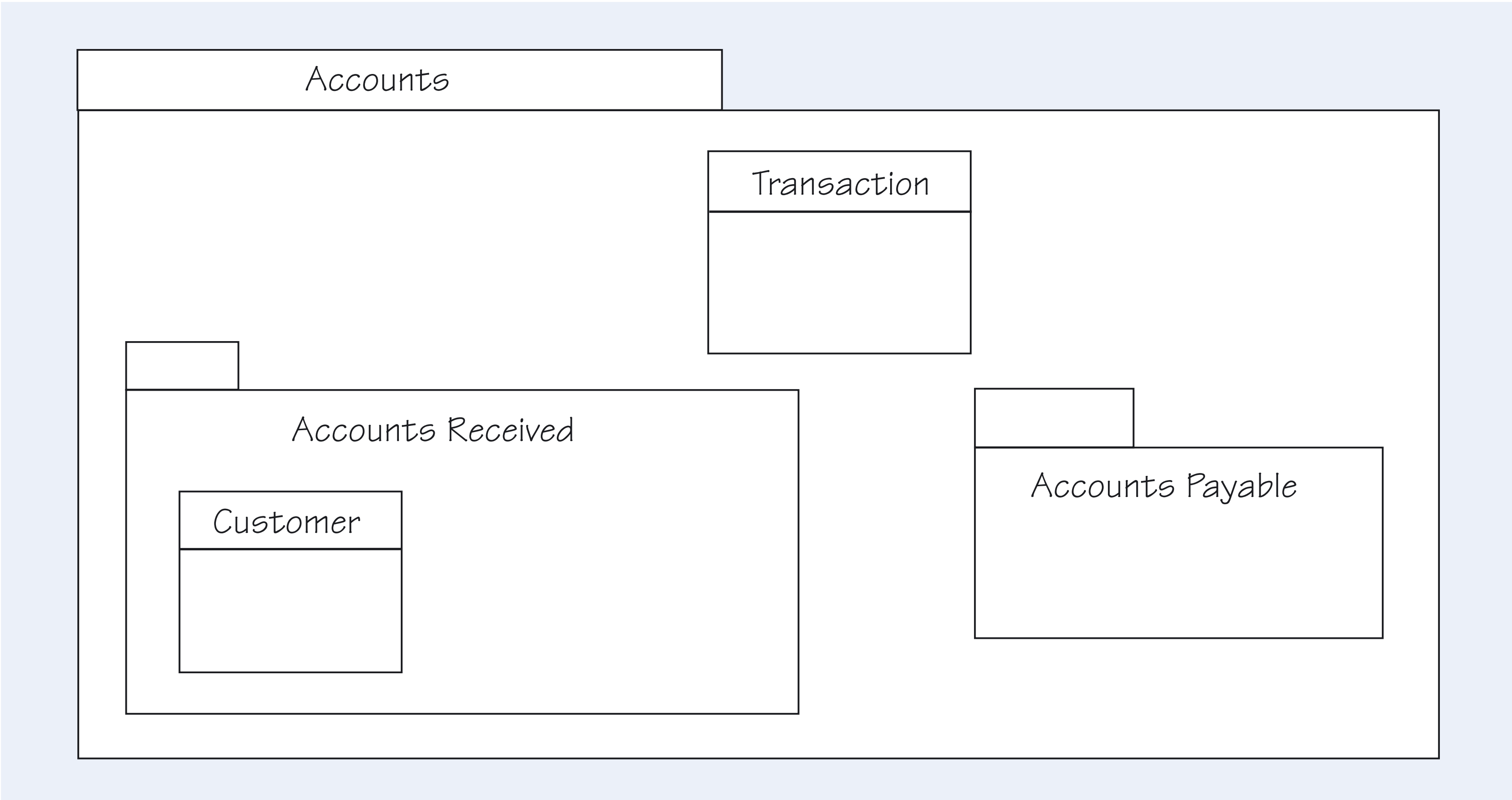


Figure 34 Packages can be nested inside each other

Package associations

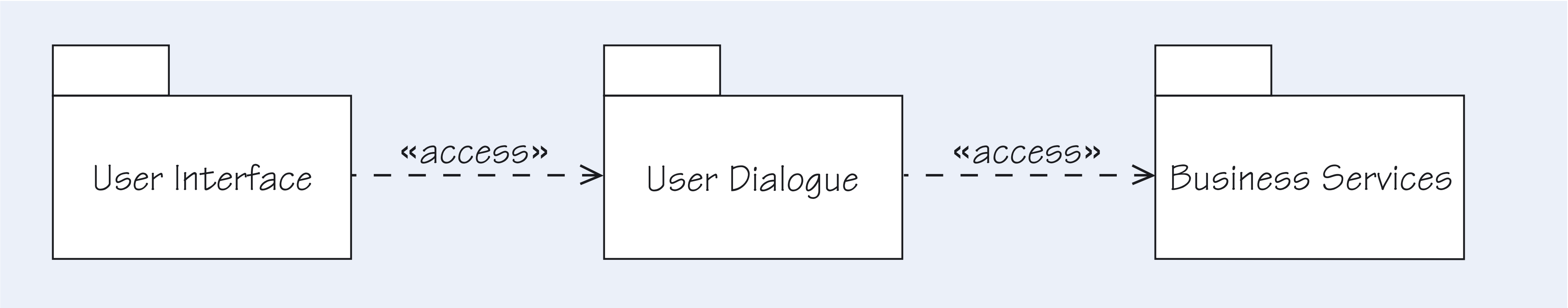


Figure 35 A three-tiered architecture

More practice questions!

IF WE HAVE TIME

Question 5

Draw a class diagram to represent the structure of the proposed system for Walton Hire

- a. Show the classes and their relationships for Walton Hire. (p19)
- b. Now add the multiplicities to the relationships (p21)
- c. Now add the attributes (p205)
- d. Add a constraint to the diagram to show past and current loans (p59)
- e. Show how this constraint could be written in OCL. (p59-69)

In this question you will find it useful to refer back to your discussion on class identification as you may have ruled out some relationships.

Question 6

Given your final diagram in Q1 part d, is it possible to derive the following, in each case briefly explain your answer.

- a. Which members have loaned a specific film?
- b. Who has seen a specific film?

Next tutorial

- Work though TMA style questions as a group.
- Identify common errors that students make.